Python Basics (1)

POLI3148 Data Science in Politics and Public Administration

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Agenda

- Section 1.
  - Why are we doing all these?
    - Why programming
    - Why Python
  - Our programming platform: Why and how
- Section 2. How to read data science readings
- Section 3. First in-class exercise

Why programming for data science

Instead of…

- **Scalability**: Working with large data
- **Precision**: Give unambiguous command to machine
- **Automation and replication**: Do the same or similar tasks
- **More tools**: A large variety of computational tools
- **Customization**: Design your most desired workflow as you want

Why programming for data science

Can Generative Artificial Intelligence replace programming?

No. At least, not yet.

- Scalability?
- Precision?
- Automation and replication?
- Variety of tools?
- Customization?

Why Python

From Wikipedia

“Python is a high-level, **general-purpose** programming language. Its design philosophy emphasizes code readability with the use of significant indentation.”

“Python consistently ranks as one of the most popular programming languages, and has gained widespread use in the machine learning community.”

Created by Guido van Rossum in 1980s

Version we are using: Python 3, released in 2008.

Why Python

Skills in great demand

- Strong data science functionality
- Easy-to-understand syntax
- “All-purpose programming language” → Easy to integrate into other components of large collaborative projects

- Vivid community provides more and more tools
- Vivid community helps to learn

Zen of Python (Tim Peters 1999)

Beautiful is better than ugly.

Explicit is better than implicit.

Simple is better than complex.

Complex is better than complicated.

Flat is better than nested.

Sparse is better than dense.

Readability counts.

Special cases aren't special enough to break the rules.

Although practicality beats purity.

Errors should never pass silently.

Unless explicitly silenced.

In the face of ambiguity, refuse the temptation to guess.

There should be one-- and preferably only one --obvious way to do it.

Although that way may not be obvious at first unless you're Dutch.

Now is better than never.

Although never is often better than right now.[d]

If the implementation is hard to explain, it's a bad idea.

If the implementation is easy to explain, it may be a good idea.

Namespaces are one honking great idea – let's do more of those!

Other Programming Languages for Data Science

- Different tools have different strengths
- Python strikes a good balance
- Want to learn others? Start with Python. Programming skills are transferrable

Programming for Data Science

Inputs

- Ideas/ Questions/ Puzzles
- Data
- Computing tools
  - Software (Python)
  - Hardware (Machine)

Outputs

- Show results of computation (in figures, tables, etc.)
- Interpret these results: description, discovery, etc.

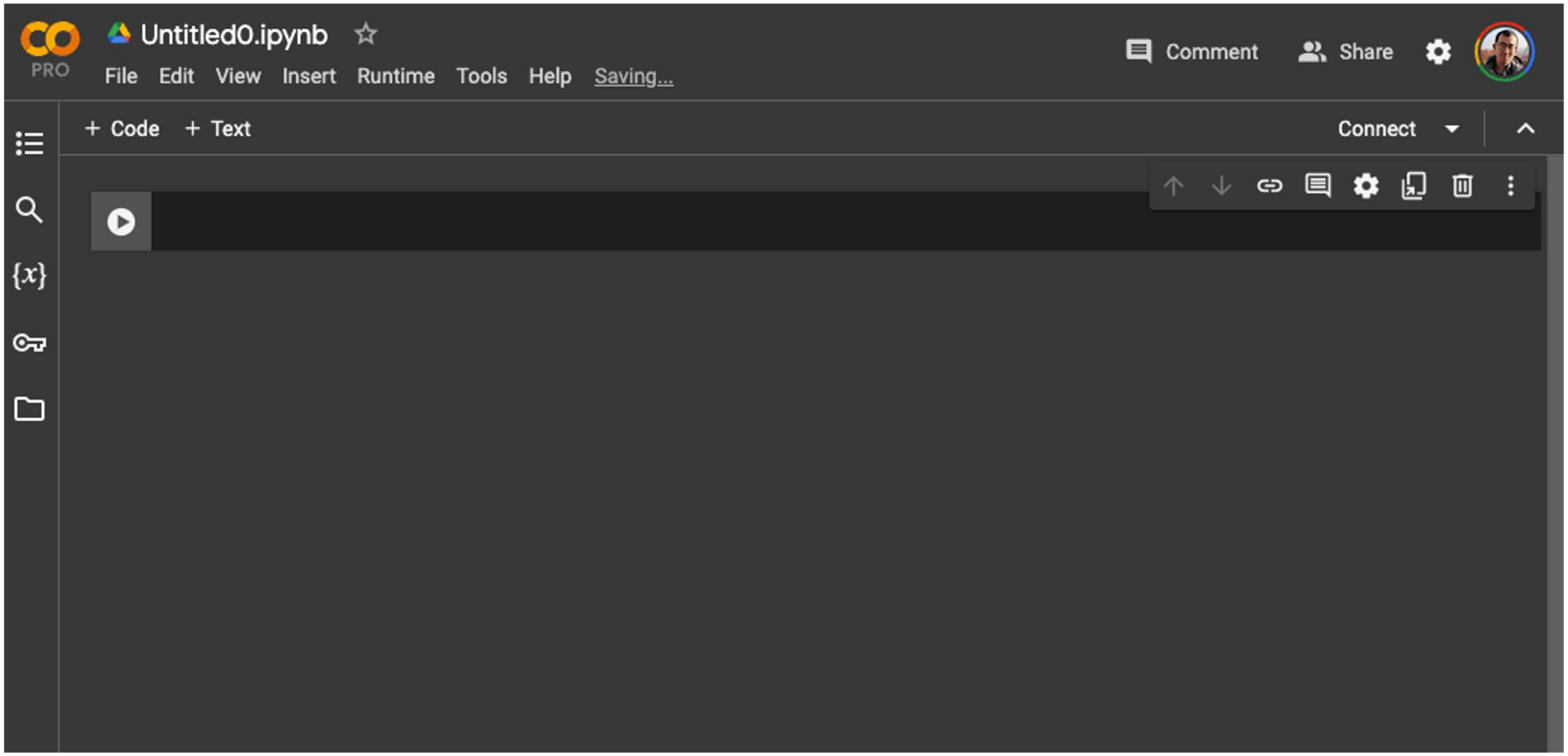
Tool: Put Inputs → Outputs Together

We need a platform that can hold all steps of data science programming

- Write down ideas
- Load the data input
- Write code to apply computing tools for data analysis
- Show results from data analysis
- Write description and discussions of the data analysis

Ultimately, you want to be able to **stably replicate** the above steps.

This is all we need



Notebooks

.ipynb file

Short for "**iPy**thon **N**ote**b**ook"

Now named "Jupyter Notebook."

Further reading:
The IPython notebook

iPython Notebook / Jupyter Notebook

The IPython Notebook, now known as the Jupyter Notebook…

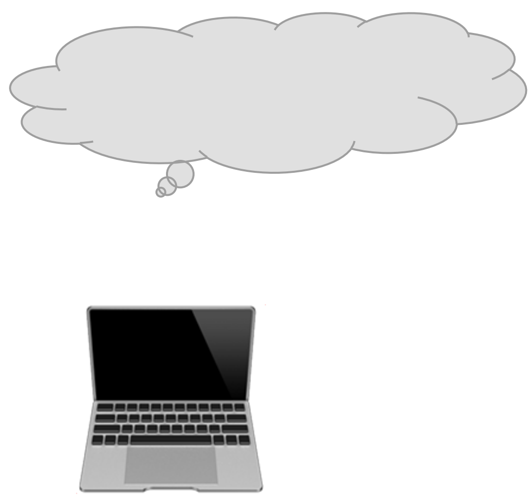
… is an **interactive** computational environment, in which you can **combine**

- code execution,
- rich text,
- mathematics,
- plots, and
- rich media.

So, what does Google Colab do?

It runs Jupyter Notebooks on Google's cloud computing infrastructure

Jupyter Notebook



Colab is a hosted Jupyter Notebook service that requires no setup to use and provides free access to computing resources, including GPUs and TPUs. Colab is especially well suited to machine learning, data science, and education.

Further reading:
https://colab.google/

Let’s do it

- Write text in the notebook
  - Section titles
  - Bullet points
  - Media
- Write code in the notebook
  - Check the status of your computing resources
  - Disconnect from/ reconnect to computing resources
- Review and commenting

Reading “Kaggle: Intro to Programming”

How to learn from your readings going forward

In this **VERY IMPORTANT** section, I would like to show you

- How to read the readings in this course
- In fact, this is also how you learn from ANY data science resources, with the least cost and highest efficiency!

PAY CLOSE ATTENTION.

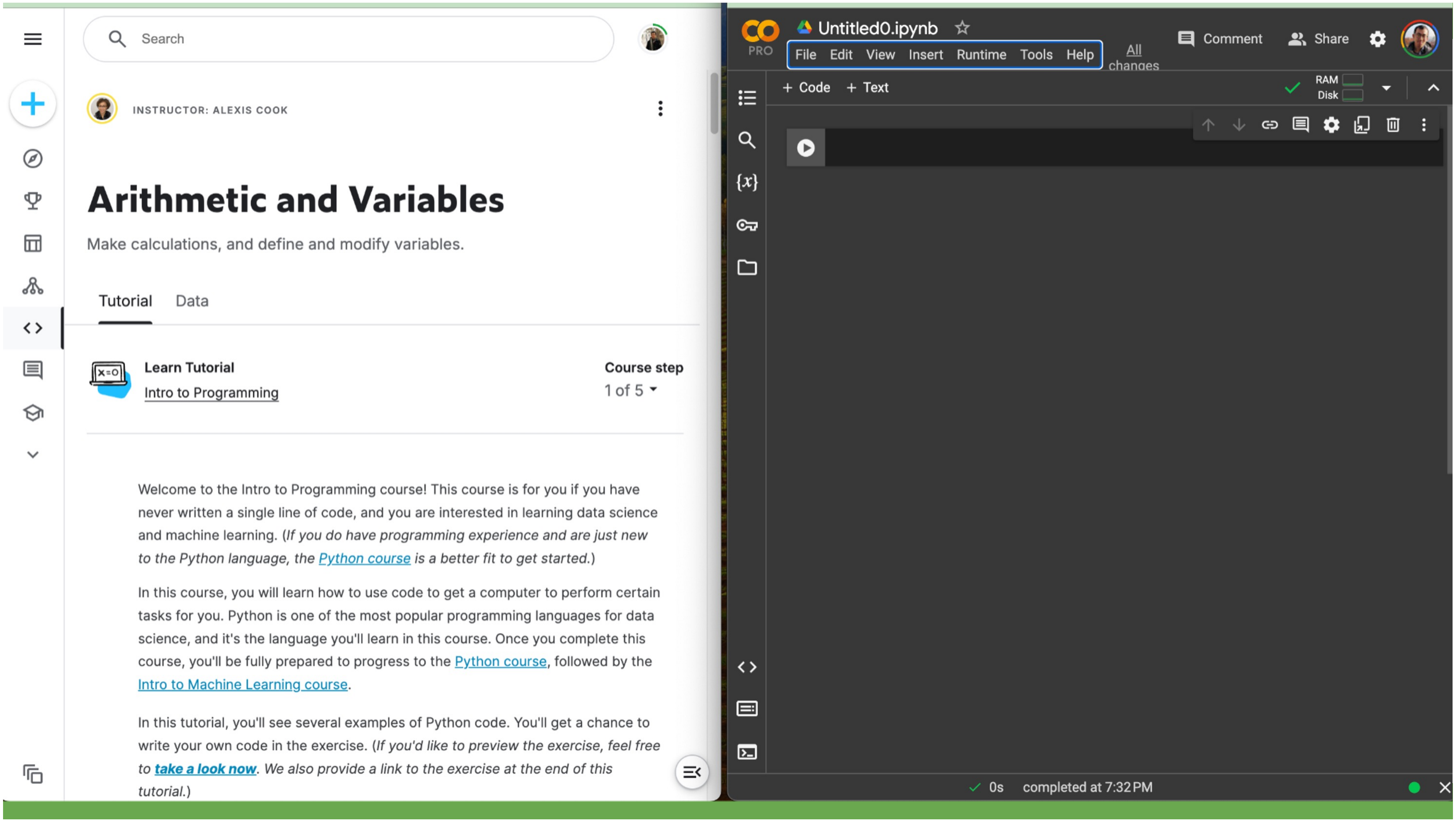
🎣

To start with...

- Open the Kaggle readings
- Set up a Jupyter Notebook on Google Colab
- Start a timer

[Optional]

- Play your favorite music 🎵
- Get yourself some comfort food 🍬🍪🍉


Search
INSTRUCTOR: ALEXIS COOK
Arithmetic and Variables
Make calculations, and define and modify variables.
Tutorial
Data
Learn Tutorial
Intro to Programming
Course step
1 of 5
Welcome to the Intro to Programming course! This course is for you if you have never written a single line of code, and you are interested in learning data science and machine learning. (If you do have programming experience and are just new to the Python language, the Python course is a better fit to get started.)
In this course, you will learn how to use code to get a computer to perform certain tasks for you. Python is one of the most popular programming languages for data science, and it's the language you'll learn in this course. Once you complete this course, you'll be fully prepared to progress to the Python course, followed by the Intro to Machine Learning course.
In this tutorial, you'll see several examples of Python code. You'll get a chance to write your own code in the exercise. (If you'd like to preview the exercise, feel free to take a look now. We also provide a link to the exercise at the end of this tutorial.)
PRO
Untitled0.ipynb
File
Edit
View
Insert
Runtime
Tools
Help
All changes
Comment
Share
+ Code
+ Text
RAM
Disk
0s
completed at 7:32 PM

To *actually* learn from data science reading

- You do it differently from how you would do for traditional reading
- Take notes!
    - Copy (type) exactly what the readings do (code)
    - Make small changes and see what happens
    - Make big changes and see what happens
    - Write down thoughts/ reminders regarding key steps
    - Document external resources you refer to

The Key: TYPE, with your hands and a keyboard

Let's do it together, now and here.

First In-Class Exercise